

SON THAI LY

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Permanent US Resident / Green Card holder

I strive to bridge the research frontier and real-world impact through creating self-optimizing AI systems that make models more efficient, adaptive, and trustworthy.

EDUCATION

University of Houston Ph.D. of Computer Engineering - Advisor: Professor Hien V. Nguyen	TX, USA 01/2021 - 12/2025
Chonnam National University Master of Computer Science	South Korea 09/2017 - 08/2019
Soongsil University Master of Material Engineering	South Korea 09/2015 - 02/2018

WORKING EXPERIENCE

University of Houston <i>Research Assistant - Machine Learning, Computer Vision, and LLMs for Medical Analysis</i>	TX, USA 06/2021 - Present
• Researched on Agentic PEFTs to optimally choose the best set of PEFT methods for given medical analysis. • Developed Be-Your-Own-Doctor, an LLM-based tool combining retrieval RAG, reasoning, and generation to generate personalized lung cancer reports (in progress). • Developed FreqFiT, a frequency-domain module that enhances PEFT performance on imaging datasets. • Led NIH project using self-supervised learning to handle limited and noisy cell annotations in brain imaging. • Published 6 papers, including 2 conference/journal accepted and 4 ArXiv papers. • Reviewer for CVPR, ICCV, MICCAI, IEEE Trans. on Circuits and Systems for Video Technology.	
Chonnam National University <i>Research Assistant - Specialized on Machine Learning, Computer Vision, Deep Learning</i>	South Korea 09/2017 - 08/2019
• Conducted research on using body gestures and 3D reconstructed facial data for <i>Expression Recognition</i>. • Published 3 international conference papers and 1 paper on Image and Vision Computing Journal.	
Soongsil University <i>Research Assistant</i>	South Korea 09/2015 - 08/2017
• Led the government R&D project on <i>4D Printing with Shape Memory Polymers</i>. • Published a paper on <i>International Journal of Precision Engineering and Manufacturing-Green Technology</i>.	

PH.D PROJECTS (06/2021 - PRESENT) | SEE [GOOGLE SCHOLAR](#)

- **Enhancing PEFT via Frequency-Domain Reweighting for Medical Imaging Analysis**, 1st author, Submitted to IEEE Transactions on Medical Imaging.
- **Frequency Strikes Back: Boosting Parameter-Efficient Foundation Model Adaptation for Medical Imaging**, 1st author, Accepted to MICCAI 2025 - **Oral presentation**. [Paper](#). [Pytorch code](#).
- **μ -Tuning - Optimizing Your Foundation Model for Medical Images: A Comprehensive Analysis of Fine-Tuning Strategies**, 1st author, [Pytorch code](#), submitted to [Medical Image Analysis](#).
- **Multiplexed Immunofluorescence Brain Image Analysis Using Self-Supervised Dual-Loss Adaptive Masked Autoencoder**, 1st author, [ArXiv](#), [Pytorch code](#), [Artificial Intelligence in Medicine](#).
- **Anomaly detection in satellite videos using diffusion models**, IEEE 26th International Workshop on Multimedia Signal Processing.

SKILLS

Frameworks	Pytorch, TensorFlow
Languages	Python, Bash, MATLAB
Tools	NumPy, Pandas, Scikit-learn, Matplotlib, Docker, Git, SLURM
Specialized	Self-supervised learning, parameter-efficient fine-tuning, CV, ML, AI, LLMs, medical imaging analysis
Design tool	AutoCAD, SolidWork, Ansys, Abaqus, Matlab Simulink